Unitree G1 locomotion Reinforcement learning in simulation

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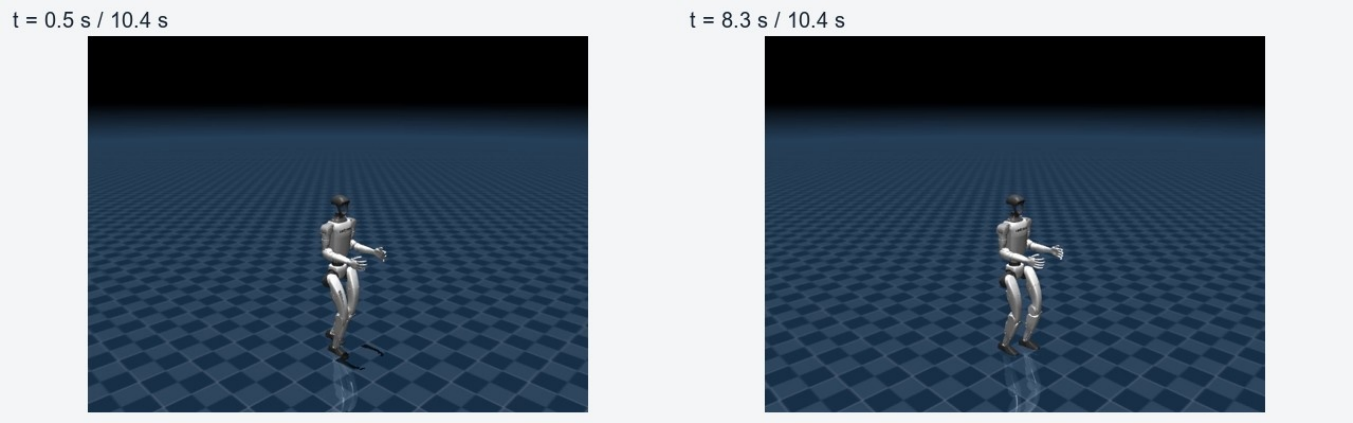
Project portfolio and video: github.com/xuhaoyan582-cell/unitree-g1-rl-portfolio

Placement context	TUS Robotics Research Laboratory, Athlone Campus
Placement supervisor	Dr Yuansong Qiao

Project focus

My placement explored reinforcement learning for humanoid locomotion using existing open-source frameworks. The practical work covered environment configuration, unsuccessful training runs and diagnostics, cloud-based training, policy export and simulation replay. Retained materials document early mjlab experimentation and a later Isaac Gym / unitree_rl_gym workflow.

A retained MuJoCo demonstration



Selected frames from the retained g1_walk.mp4 recording. The clip shows a G1 simulation replay; camera tracking follows the robot. Frame timestamps refer to the encoded video.

Recorded test	One 10.0-second simulation run; 5,000 simulation steps
Forward command	0.5 m/s
Terminal position	x = 4.081 m; y = 0.296 m; base z = 0.756 m
Console outcome	The recording script reported: Robot stayed upright

The values above are transcribed from the historical console record linked to this recording. They describe one demonstration, rather than a repeated robustness evaluation.

METHOD, CONTRIBUTION AND SUPPORTING MATERIAL

From training to simulation replay

The retained evidence connects a recorded cloud-training checkpoint, an exported recurrent policy and a local MuJoCo replay. This summary focuses on that documented workflow.

Workflow retained in the project record

1 / Train and export	The historical console record loads model_5950.pt from a unitree_rl_gym G1 run and exports policy_lstm_1.pt.
2 / Replay a selected policy	The recording command explicitly selects the exported policy file for a MuJoCo replay with a forward command of 0.5 m/s.
3 / Retain outputs	The local archive contains the exported policy, G1 configuration, EGL recording script and g1_walk.mp4.

Practical work and learning

I worked with existing frameworks to configure the simulation and cloud workflow, inspect training behaviour, troubleshoot environment compatibility and recording, and organise a trained policy for replay. Early mjlab records show custom task registration, configuration changes and training diagnostics; the retained successful demonstration belongs to the later Isaac Gym workflow.

Technical snapshot

Training / replay stack	Isaac Gym and unitree_rl_gym; recurrent policy exported for MuJoCo replay
Retained policy interface	47 observation inputs; LSTM hidden size 64; 12 action outputs, consistent with the local G1 configuration
Recorded environment	Historical installation logs list Isaac Gym 1.0rc4, PyTorch 1.13.1+cu117 and MuJoCo 3.2.3.

Supporting materials

Demonstration: retained g1_walk.mp4, approximately 10.4 seconds of encoded video.
Implementation snapshot: G1 configuration, exported recurrent policy and headless EGL recording script.
Process record: archived console records dated 26 April 2026 (UTC), including checkpoint loading, export and the recording result.
Placement context: existing placement report and poster; supervisor confirmation may be requested to verify the placement and personal responsibilities.

Evaluation scope and next steps

The demonstrated result is simulation-based. Repeated trials across command speeds, initial states and disturbances would be needed to quantify reliability and generalisation. Future work would also document evaluation metrics, retain complete run logs and assess requirements for physical deployment.

Evidence note: this portfolio retains an exported policy; the original model_5950.pt checkpoint and full training event logs are not included. The checkpoint-to-export link is supported by the archived console record. This is an implementation snapshot, rather than a complete reproducibility package.